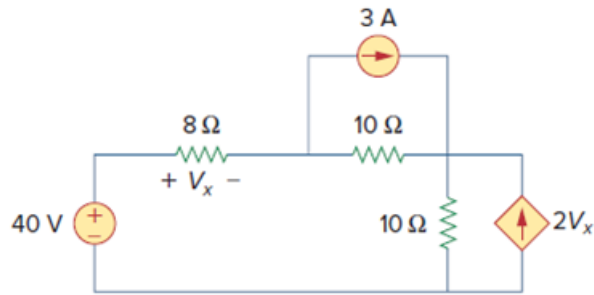


Problem

Use source transformation to find the voltage V_x in the circuit of Fig. 4.92.

Figure 4.92:

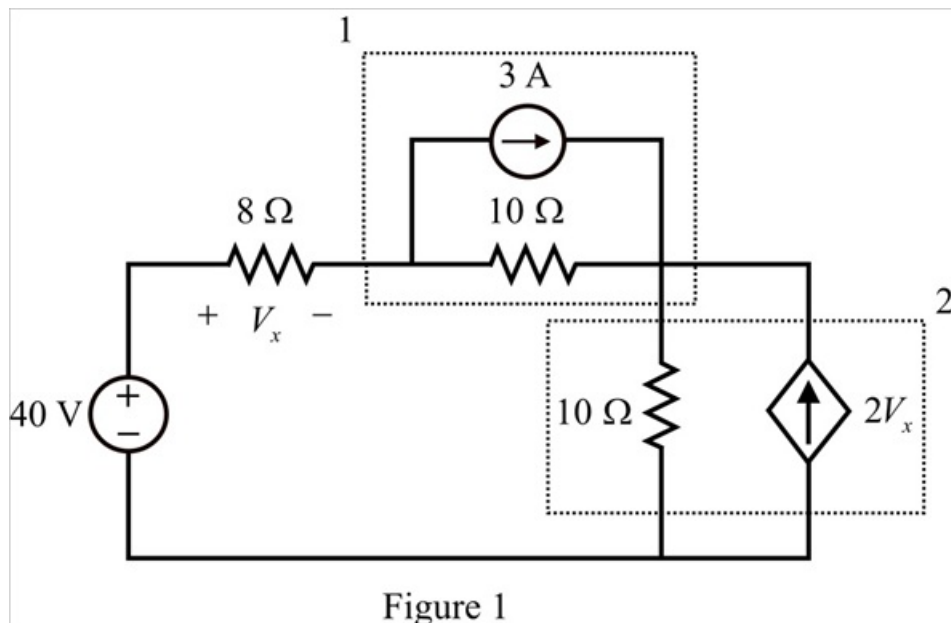


Step-by-step solution

Step 1 of 5

Refer to the circuit diagram depicted in Figure 4.92 in the textbook.

Draw the modified circuit diagram shows with the branches that are needed to be transformed.



Step 2 of 5

In Figure 1, there are two parallel branches with current source and resistors. Transform the combination 1, in which 3 A current source is in parallel with $10\ \Omega$ resistor into the series combination of equivalent voltage source and $10\ \Omega$ resistor. The value of equivalent voltage source is,

$$\begin{aligned} V &= (3)(10) \\ &= 30\text{ V} \end{aligned}$$

Similarly, transform the combination 2, in which $2V_x$ dependent source is in parallel with $10\ \Omega$ resistor into the series combination of equivalent dependent voltage source and $10\ \Omega$ resistor. The value of equivalent dependent voltage source is,

$$\begin{aligned} V &= (2V_x)(10) \\ &= 20V_x\text{ V} \end{aligned}$$

Step 3 of 5

Draw the modified circuit diagram.

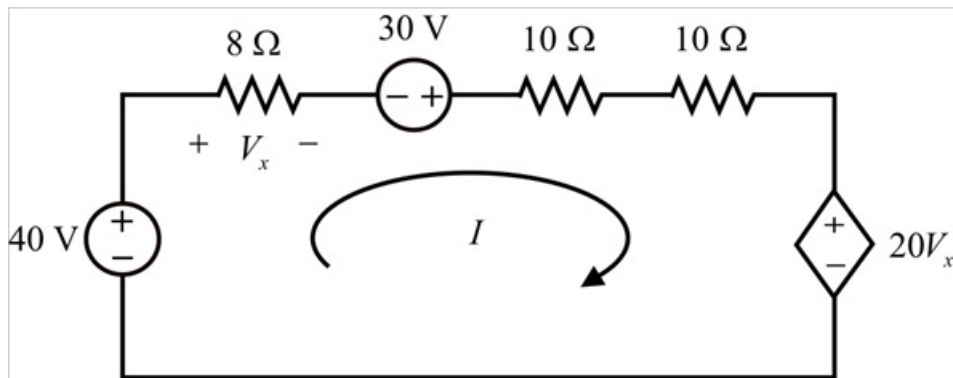


Figure 2

Step 4 of 5

From the Figure 2, the value of voltage V_x is,

$$V_x = 8I$$

Apply Kirchhoff's voltage law around the loop in Figure 2.

$$-40 + 8I - 30 + 10I + 10I + 20V_x = 0$$

$$28I + 20V_x = 70$$

$$28I + 20(8I) = 70 \quad (\text{since } V_x = 8I)$$

$$188I = 70$$

$$I = 0.372\text{ A}$$

Step 5 of 5

Calculate the voltage V_x .

$$V_x = 8I$$

$$= 8(0.372)$$

$$= 2.98\text{ V}$$

Therefore, the value of voltage V_x in the circuit is $\boxed{2.98\text{ V}}$.